Geodesic-Pancyclic Graphs

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Abstract

A shortest path connecting two vertices u and v is called a u-v geodesic. The distance between u and v, denoted by $d_G(u,v)$, is the number of edges in a u-v geodesic. A graph G with n vertices is geodesic-pancyclic if for each pair of vertices $u,v \in V(G)$, every u-v geodesic lies on every cycle of length k satisfying $max\{2d_G(u,v),3\} \leq k \leq n$. In this paper, we study the properties for graphs to be geodesic-pancyclic. In particular, we show that some sufficient conditions for panconnected graphs still suffice for geodesic-pancyclic graphs.

Keyword: Hamiltonian; Panconnected; Pancyclic; Vertex-pancyclic; Edge-pancyclic; Geodesic-pancyclic

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1 Introduction

All graphs considered in this paper are finite and simple (i.e., without loops and multiple edges). We denote the vertex set and edge set of a graph G by $V(G)$ and $E(G)$, respectively. A graph G is *k-connected* if the removal of $k-1$ or fewer vertices leaves neither a disconnected graph nor a trivial one. For a vertex $u \in V(G)$, we denote by $N_G(u)$ the set consisting of all vertices adjacent to u in G, and let $deg_G(u) = |N_G(u)|$ denote its degree. When no ambiguity arises, we omit the subscript G in the above notations. Also, we define $\delta(G) = \min\{deg(u) : u \in V(G)\}$ and $\sigma_2(G) = \min\{deg(u) + deg(v) : u,v \in V(G) \text{ and } uv \notin E(G)\}$. For two vertices $u,v \in V(G)$, a path P joining u and v is called a *u-v path*. In particular, if P is a shortest path, the path is called a *u-v geodesic*. The *distance* between u and v, denoted by $d_G(u,v)$, is the number of edges in a u-v geodesic. The *diameter* of a graph G, denoted by $diam(G)$, is $\max\{d_G(u,v) : u,v \in V(G)\}$. For a subset $S \subset V(G)$, we use $G - S$ to denote the graph obtained from G by removing S along with its incident edges. If H is a subgraph of G, then for simplicity, we sometimes write H to mean $V(H)$.

In the following definitions, we assume that a graph G is connected with order n (i.e., $|V(G)| = n$). A path (respectively, cycle) that contains every vertex of G exactly once is called a *hamiltonian path* (respectively, *hamiltonian cycle*). A graph G is *traceable* (respectively, *hamiltonian*) if it possesses a hamiltonian path (respectively, hamiltonian cycle). Several hamiltonian-like properties have been widely investigated in the literatures (see [5] for a recent survey). A graph G is *hamiltonian-connected* if every two vertices of G are connected by a hamiltonian path. A graph G is *panconnected* if, for each pair of vertices $u, v \in V(G)$ and for each integer k satisfying $d_G(u, v) \leq k \leq n-1$, there is a u-v path of length k in G. For $r \geq 3$ an integer, a graph G is called *r-pancyclic* if it contains a cycle of length k (i.e., a k-cycle) for every k between r and n. Furthermore, G is called *vertex r-pancyclic* (respectively, *edge r-pancyclic*) if every vertex (respectively, edge) of G belongs to a k-cycle for each $k = r, \ldots, n$. A 3-pancyclic graph, a vertex 3-pancyclic graph and an edge 3-pancyclic graph are called pancyclic, vertex-pancyclic and edge-pancyclic, respectively. A graph G is *k-distant hamiltonian*, $0 \leq k \leq diam(G)$, if whenever u and v are vertices of G for which $d_G(u, v) \leq k$, there is a hamiltonian cycle C of G such that $d_C(u, v) = d_G(u, v)$.

The following hamiltonian-like definitions are first introduced in this paper. For a nonnegative integer s, a pair of vertices $\langle u, v\rangle$ is said to be *geodesic s-pancyclic* if every u-v geodesic lies on every cycle of length k satisfying $max\{2d_G(u, v) + s, 3\} \leq k \leq n$. A graph G is *geodesic s-pancyclic* if, for every two vertices $u, v \in V(G)$, $\langle u, v\rangle$ is geodesic s-pancyclic. In particular, the class of geodesic 0-pancyclic graphs is simply called *geodesic-pancyclic*, which is properly contained in the class of edge-pancyclic graphs and the class of λ-distant hamiltonian graphs, where $\lambda = diam(G)$. In general, every geodesic s-pancyclic graph must be edge $(s+2)$-pancyclic if $s \geq 1$. Also, note that every geodesic s-pancyclic graph is geodesic t-pancyclic if it provides $s \leq t$.

There are several well-known and important results for panconnected graphs [1, 2, 9]. In this paper, we shall prove that most of sufficient conditions for panconnected graphs still suffice for geodesic-pancyclic graphs. However, a graph G which is panconnected does not imply it is geodesic-pancyclic. The wheel W_n of order n is the graph obtained by joining each vertex of the $(n-1)$-cycle to a common vertex different from those on the cycle. The wheels W_n with $n \geq 4$ mentioned in [9] are panconnected. An easy observation shows that W_n with $n \geq 5$ are not geodesic-pancyclic. For example, Figure 1 shows that the v_1-v_3 geodesic v_1, v_0, v_3 does not lie on any 5-cycle in W_5. Contrasting the properties for graphs to be panconnected and/or edge-pancyclic, thereafter we study the characterizations for graphs to be geodesic-pancyclic.

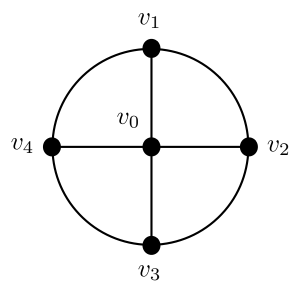

Figure 1: W_5

2 A necessary condition

We start with a lemma that proves a necessity of geodesic 1-pancyclic graphs.

Lemma 1 *If G is a geodesic 1-pancyclic graph of order $n \geq 4$, then G is 3-connected.*

Proof. Recall that if G is geodesic 1-pancyclic then G is edge-pancyclic, and this further implies that G is 2-connected. We first show that every vertex $w \in V(G)$ has at least three neighbors. By the fact $\delta(G) \geq 2$, we let $u, v \in N_G(w)$. Consider either $uv \in E(G)$ or $uv \notin E(G)$ as follows. If $uv \in E(G)$, we let C be a hamiltonian cycle passing through uv in G. Since $n \geq 4$, at most one vertex of u and v can be adjacent to w in C. This shows that there exists another vertex $z \in G - \{u, v, w\}$ such that it is adjacent to w in C. On the other hand, if $uv \notin E(G)$, since uw lies on a 3-cycle in G, it guarantees that there is a vertex $z \in G - \{u, v, w\}$ such that it is adjacent to both u and w in G. Hence, $\delta(G) \geq 3$.

To show that G is 3-connected, we suppose the contrary that there exist two vertices $u, v \in V(G)$ such that $G - \{u, v\}$ is a disconnected graph. We again consider either $uv \in E(G)$ or $uv \notin E(G)$. For the former case, since $G - \{u, v\}$ is disconnected, uv does not lie on any n-cycle in G. This contradicts to the fact that G is edge-pancyclic. For the latter case, let P be any u-v geodesic in G. It is clear that $P - \{u, v\}$ must be contained in one connected component of $G - \{u, v\}$, say K. Since $G - \{u, v\}$ is disconnected, we assume that there is a vertex $x \notin K$. Moreover, since u and v are nonadjacent and $\delta(G) \geq 3$, there is a vertex $w \in P - \{u, v\}$ such that w is adjacent to a vertex $z \in K - P$. Thus, x and z are contained in different components of $G - P$. From the fact that $G - P$ is disconnected, P cannot lie on any n-cycle in G, a contradiction. □

3 Dirac's and Ore's conditions

Dirac [3] first proposed a sufficient condition for a graph to be hamiltonian in 1952. Successively, Ore [6] generalized Diracs condition by looking at the degree sum of two nonadjacent vertices and purposed an attractive improvement.

Theorem 2 [3] *If G is a graph of order $n \geq 3$ such that $\delta(G) \geq n/2$, then G is hamiltonian.*

Theorem 3 [6] *If G is a graph of order $n \geq 3$ such that $\sigma_2(G) \geq n$, then G is hamiltonian.*

Subsequent papers by several authors have shown that these two types of sufficient condition for restricted classes of hamiltonian graphs. As shown in Theorem 4, Randerath et al. [8] offered the best possible Dirac's condition for graphs to be edge-pancyclic. In fact, in a very early time, Williamson [9] has obtained a more qualified result showing graphs which are panconnected under the same condition.

Theorem 4 [8] *If G is a graph of order $n \geq 4$ such that $\delta(G) \geq (n + 2)/2$, then G is edge-pancyclic.*

Theorem 5 [9] *If G be a graph of order $n \geq 4$ such that $\delta(G) \geq (n + 2)/2$, then G is panconnected.*

Analogously, we shall present the best possible Dirac's condition for graphs to be geodesic-pancyclic. Before this, we recall that Faudree and Schelp [4] have provided the following condition for graphs to be 4-panconnected (i.e., a panconnected graph without 3-cycle).

Theorem 6 [4] *Let G be a graph of order $n \geq 5$ such that $\sigma_2(G) \geq n+1$. Then every pair of distinct vertices of G are connected by a path of length k for each $k = 4, \ldots, n-1$.*

Theorem 7 *Let G be a graph of order $n \geq 4$ such that $\delta(G) \geq (n+2)/2$. Then G is geodesic-pancyclic.*

Proof. It is easy to verify that G is a complete graph whenever $n = 4$ or $n = 5$ and $\delta(G) \geq (n+2)/2$. Here we consider only for $n \geq 6$. Let u, v be any two distinct vertices of G. Since $\delta(G) \geq (n+2)/2$, u and v have at least two common neighbors in G and hence $d_G(u,v) \leq 2$. If $d_G(u,v) = 1$, then $\langle u,v \rangle$ is geodesic-pancyclic by Theorem 4.

We now consider $d_G(u,v) = 2$. Since $|N_G(u) \cap N_G(v)| \geq 2$, every u-v geodesic is contained in a 4-cycle in G. Let $w \in N_G(u) \cap N_G(v)$ and $H = G - w$. For proving $\langle u,v \rangle$ is geodesic-pancyclic in G, it suffices to show that there exist paths of every length $k = 3, \ldots, n-2$ between u and v in H. Then, these paths together with uwv produce the desired cycles containing a u-v geodesic in G. Since $|N_G(u) \cap N_G(v)| \geq 2$, we have $d_H(u,v) = 2$. We first claim that there is a path of length 3 connecting u and v in H. Suppose not and let x be any vertex adjacent to u in H. Then, for any $y \in N_H(v)$, x and y are nonadjacent in G. Since $deg_H(v) = deg_G(v) - 1 \geq n/2$, it implies $deg_G(x) \leq (n-1) - n/2 \leq n/2$, a contradiction. Moreover, since $deg_H(u) \geq n/2$ and $deg_H(v) \geq n/2$, as well as u and v are arbitrarily chosen from G with distance 2 apart in H, we conclude that $\sigma_2(H) \geq (n/2) + (n/2) = n$ (i.e., the order of H plus one). By Theorem 6, u and v in H are connected by a path of length k for each $k = 4, \ldots, n-2$. Thus $\langle u,v \rangle$ is geodesic-pancyclic and we are done. $\square$

The result of Theorem 7 is the best possible in the sense that the hypothesis cannot be weakened. Consider the family of graphs $\{J_n : n \geq 7 \text{ is an odd integer}\}$ defined by $J_n = (\frac{n-1}{2})K_1 + (K_2 \cup K_{\frac{n-3}{2}})$ (See Figure 2 for visual illustrations). We can verify that $\delta(J_n) = (n+1)/2$ and the edge uv does not lie on any 4-cycle. This shows that J_n is not geodesic-pancyclic.

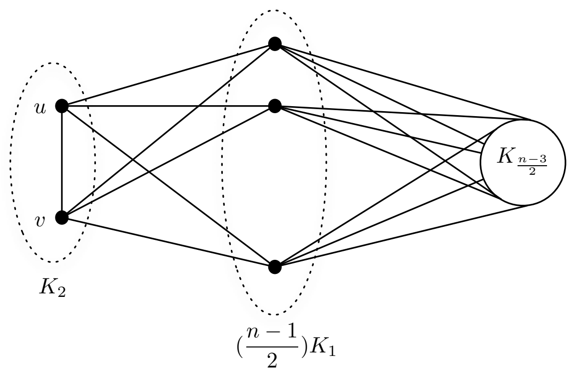

Figure 2: J_n

In view of the Ore's condition, Williamson [9] has proved out the best possible condition for graphs to be panconnected. In the following, we show that we can obtain the best possible Ore's condition for graphs to be geodesic-pancyclic under the same hypothesis of Theorem 8.

Theorem 8 [9] *If G is a graph of order $n \geq 4$ such that $\sigma_2(G) \geq (3n-2)/2$, then G is panconnected.*

Theorem 9 *Let G be a graph of order $n \geq 4$ such that $\sigma_2(G) \geq (3n-2)/2$. Then G is geodesic-pancyclic.*

Proof. We suppose the contrary that G is not geodesic-pancyclic. By Theorem 7, the graph G contains a vertex u such that $deg(u) < (n+2)/2 \leq n-1$ when $n \geq 4$. This implies that there exists a vertex v in G which is not adjacent to u. Thus $deg(v) \leq n-2$ and $deg(u) + deg(v) < (n+2)/2 + (n-2) = (3n-2)/2$ which is a contradiction. Therefore, G must be geodesic-pancyclic. □

The bound in the above theorem is sharp. Consider the family of graphs $\{L_n : n \geq 4 \text{ is an even integer}\}$ defined as follows. L_n consists of two adjacent vertices u and v together with a clique K_{n-2} whose vertices are partitioned into two disjoint sets X and Y such that u is adjacent to every vertices of X and v is adjacent to every vertices of Y. As shown in Figure 3, we are easy to check $deg(u) = n/2$ and $deg(y) = n-2$ for every $y \in Y$. Thus $\sigma_2(L_n) = deg(u) + deg(y) = (3n-4)/2$. Since uv does not lie on any 3-cycle, L_n is not geodesic-pancyclic.

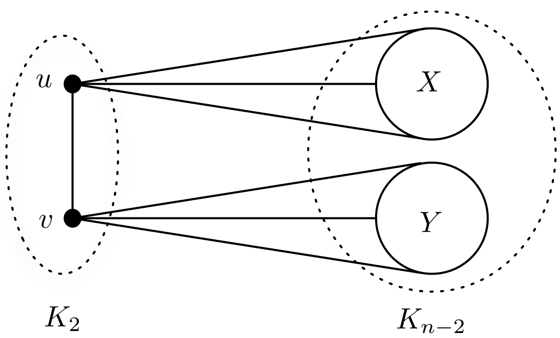

Figure 3: L_n

4 Extreme size conditions

Throughout this sction, we denote m_G as the size (i.e., the number of edges) of G. Ore [7] and Williamson [9] determined the best possible lower bound of size for graphs to be hamiltonian and panconnected, respectively.

Theorem 10 [7] *If G is a graph of order $n \geq 3$ and size $m_G \geq \binom{n-1}{2} + 2$, then G is hamiltonian.*

Theorem 11 [9] *If G is a graph of order $n \geq 4$ and size $m_G \geq \binom{n-1}{2} + 3$, then G is panconnected.*

Corollary 12 *If G is a graph of order $n \geq 3$ and size $m_G \geq \binom{n-1}{2} + 1$, then G is traceable.*

Proof. The proof is by induction on n. It is easy to verify for $n = 3, 4$. Suppose that the corollary is true for all graphs with order less than n. We now consider G to be a graph with n vertices that satisfies the hypothesis of the corollary. Let v be a vertex in G such that $deg(v) = \delta(G)$ and let $H = G - v$. Since $m_H \leq \binom{n-1}{2}$, $deg(v) = m_G - m_H \geq 1$. If $deg(v) = n-1$, then G is a complete graph and thus is traceable. Otherwise, we consider the following two cases where $p = n/2$ for n even and $p = (n-1)/2$ for n odd.

Case 1. $1 \leq deg(v) \leq p$. Since $m_H = m_G - deg(v) \geq \binom{n-1}{2} + 1 - p$, we obtain $m_H \geq \binom{n-2}{2} + 2$ whenever $n \geq 5$ is odd or $n \geq 6$ is even. By Theorem 10, H is hamiltonian. Let $C = v_0, v_1, \ldots, v_{n-2}, v_0$ be a hamiltonian cycle in H. Since $deg(v) \geq 1$, v is adjacent to at least one vertex of C, say v_i. Thus, $v, v_i, v_{i+1}, \ldots, v_{n-2}, v_0, \ldots, v_{i-1}$ forms a hamiltonian path in G.

Case 2. $p+1 \leq deg(v) \leq n-2$. Clearly, $m_H = m_G - deg(v) \geq \binom{n-1}{2} + 1 - (n-2) = \binom{n-2}{2} + 1$. From the induction hypothesis, we know that H is traceable. Let $P = v_0, v_1, \ldots, v_{n-2}$ be a hamiltonian path in

H. Since $deg(v) \geq p+1$, there exist two consecutive vertices v_i and v_{i+1} in P that are adjacent to v. Thus, $v_0, v_1, \ldots, v_i, v, v_{i+1}, \ldots, v_{n-2}$ forms a hamiltonian path in G. □

We now show that graphs are geodesic-pancyclic under the same condition of Theorem 11.

Theorem 13 *Let G be a graph of order $n \geq 4$ and size $m_G \geq \binom{n-1}{2}+3$. Then G is geodesic-pancyclic.*

Proof. Let u, v be any two vertices of G. We first claim that $d_G(u,v) \leq 2$. To see this, we let q be the number of edges incident with at least one of u or v in G and let $H = G - \{u,v\}$. Clearly, $m_H \leq \binom{n-2}{2}$. Thus, $q = m_G - m_H \geq \binom{n-1}{2} + 3 - \binom{n-2}{2} = n+1$. This implies that u and v have a common neighbor in G and thus $d_G(u,v) \leq 2$. Hence, we only need to consider $d_G(u,v) = 1$ or 2. If $d_G(u,v) = 1$, then $\langle u, v\rangle$ is geodesic-pancyclic by Theorem 11.

We now consider $d_G(u,v) = 2$. In this case, $q = deg_G(u) + deg_G(v) \leq 2n-4$. Let $w \in N_G(u) \cap N_G(u)$ and $G' = G - w$. For proving that $\langle u, v\rangle$ is geodesic-pancyclic, it suffices to show that there exist paths of length from 2 to $n-2$ connecting u and v in G'. The result is obvious for $n=4$ or 5 since the former case indicates that G is a complete graph and the latter case indicates that G is a complete graph by deleting any one edge. Let $H' = G - \{u,v,w\}$ and $q' = m_G - m_{H'}$. Since $deg_G(w) \leq n-1$, $q' = deg_G(u) + deg_G(v) + deg_G(w) - 2 = q + deg_G(w) - 2 \leq q + n - 3$. We further discuss the following two cases with $n \geq 6$.

Case 1. $q \leq 2n-5$. Since $q' \leq q+n-3 \leq 3n-8$, we have $m_{H'} = m_G - q' \geq \binom{n-1}{2} + 3 - (3n-8) = \binom{n-4}{2} + 2$. By Theorem 10, H' is hamiltonian. Let $C = v_0, v_1, \ldots, v_{n-4}, v_0$ be a hamiltonian cycle in H'. Suppose that G' does not contain a path of length k between u and v for some $2 \leq k \leq n-2$. Then, for every vertex $v_i \in C$ adjacent to u, the vertex v_j with $j \equiv i+(k-2) \pmod{n-3}$ cannot be adjacent to v in G'. Thus $|N_{G'}(u)| + |N_{G'}(v)| \leq n-3$. However, since $q = deg_G(u) + deg_G(v) \geq n+1$, it implies $|N_{G'}(u)| + |N_{G'}(v)| = (deg_G(u) - 1) + (deg_G(v) - 1) \geq n-1$, a contradiction. Thus, G' contains u-v paths of length k for all $2 \leq k \leq n-2$.

Case 2. $q = 2n-4$. Using an argument similar to Case 1 we can show that $m_{H'} = m_G - q' \geq \binom{n-4}{2} + 1$. By Corollary 12, H' is traceable. Let P be a hamiltonian path in H'. Since $q = 2n-4$, u and v must be adjacent to every vertex of P. Thus, we can easily construct u-v paths of length k in G' for every $2 \leq k \leq n-2$ in an obvious way. □

Let R_n be a graph consisting of a clique K_{n-1} together with a vertex w such that w is adjacent to exact two vertices of K_{n-1}, say u and v (See Figure 4 for visual illustrations). Clearly, R_n has exactly $\binom{n-1}{2}+2$ edges. Since uv does not lies on any n-cycle, R_n is not geodesic-pancyclic. Thus, the family of graphs $\{R_n : n \geq 4 \text{ is an integer}\}$ shows that Theorem 13 is the best possible.

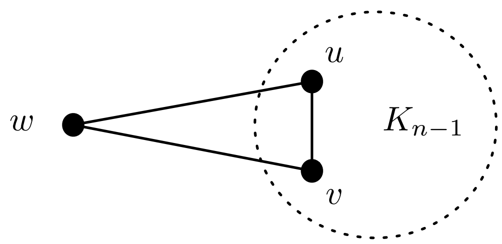

Figure 4: R_n

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